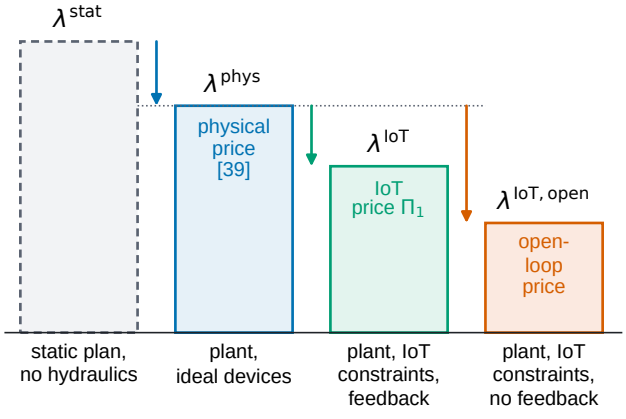
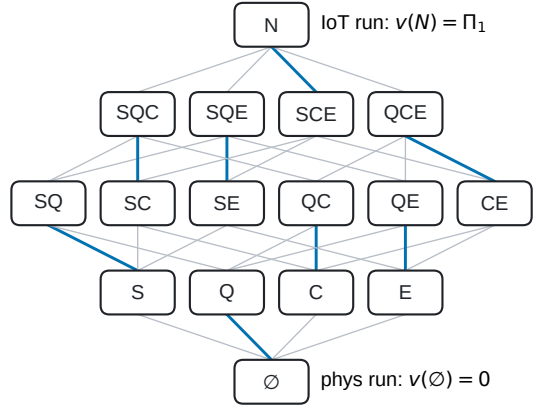
(a)

Schematic; levels and ordering are illustrative, not data.



(b)

$v(F)$: mean daily Π_1 with only the constraints in F active



adds Q: marginal contribution $v(F \cup \{Q\}) - v(F)$, averaged into ϕ_Q

$$\phi_i = \sum_{F \subseteq N \setminus \{i\}} \frac{|F|!\,(3-|F|)!}{4!} [v(F \cup \{i\}) - v(F)], \quad \sum_i \phi_i = \Pi_1$$